DROUGHT, HEAT WAVES AND SLOW-ONSET RISK

DROUGHT IDENTITY FIREWALL → DROUGHT-FAMILY MATRIX → SLOW-ONSET CASCADE → DECLARATION DASHBOARD → HEAT HAZARD-TO-HARM CHAIN → WET-BULB FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

★ PRIMARY CORE FLOW · ◉ SUBORDINATE EXTRA · ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00

STAGE 00

DROUGHT IDENTITY FIREWALL

DROUGHT IDENTITY FIREWALL · PERSISTENT CLIMATIC MOISTURE CONDITION · TEMPORARY DEFICIT RELATIVE TO LOCAL EXPECTATION · SUPPLY-DEMAND-ACCESS CONDITION · SERIOUS DISRUPTION BEYOND COPING CAPACITY

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
ARIDITY → persistent climatic moisture condition	SCARCITY → supply-demand-access condition	ARIDITY → persistent climatic moisture condition
DROUGHT → temporary deficit relative to local expectation	DISASTER → serious disruption beyond coping capacity	DROUGHT → temporary deficit relative to local expectation

HAZARD / ACTOR / INSTRUMENT

- ARIDITY → persistent climatic moisture condition
- DROUGHT → temporary deficit relative to local expectation

DECISIVE RISK OR MANDATE DISTINCTION

- SCARCITY → supply-demand-access condition
- DISASTER → serious disruption beyond coping capacity

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- ARIDITY → persistent climatic moisture condition
- DROUGHT → temporary deficit relative to local expectation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DROUGHT → temporary deficit relative to local expectation.

01

STAGE 01

DROUGHT-FAMILY MATRIX

DROUGHT-FAMILY MATRIX · SOIL MOISTURE · CROP WATER · RIVERS RESERVOIRS LAKES AQUIFERS

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
METEOROLOGICAL → precipitation	HYDROLOGICAL → rivers reservoirs lakes aquifers	METEOROLOGICAL → precipitation
AGRICULTURAL → soil moisture / crop water	SOCIOECONOMIC / ECOLOGICAL → services-livelihoods / ecosystems	AGRICULTURAL → soil moisture / crop water

HAZARD / ACTOR / INSTRUMENT

- METEOROLOGICAL → precipitation
- AGRICULTURAL → soil moisture / crop water

DECISIVE RISK OR MANDATE DISTINCTION

- HYDROLOGICAL → rivers reservoirs lakes aquifers
- SOCIOECONOMIC / ECOLOGICAL → services-livelihoods / ecosystems

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- METEOROLOGICAL → precipitation
- AGRICULTURAL → soil moisture / crop water

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HYDROLOGICAL → rivers reservoirs lakes aquifers; SOCIOECONOMIC / ECOLOGICAL → services-livelihoods / ecosystems.

02

STAGE 02

SLOW-ONSET CASCADE

SLOW-ONSET CASCADE · RAINFALL DEFICIT · SOIL-MOISTURE STRESS · GROUNDWATER DECLINE · CROP ECOSYSTEM LIVELIHOOD AND HEALTH EFFECTS · LAGS DIFFER · MONITOR EACH STAGE

RAINFALL DEFICIT → SOIL-MOISTURE STRESS

STORAGE / GROUNDWATER DECLINE → CROP ECOSYSTEM LIVELIHOOD AND HEALTH EFFECTS → LAGS DIFFER → MONITOR EACH STAGE

RISK CONDITION / INPUT

- RAINFALL DEFICIT → SOIL-MOISTURE STRESS
- STORAGE / GROUNDWATER DECLINE

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- CROP ECOSYSTEM LIVELIHOOD AND HEALTH EFFECTS
- LAGS DIFFER → MONITOR EACH STAGE

LAST-MILE / INCLUSION / OUTCOME LIMIT

- RAINFALL DEFICIT → SOIL-MOISTURE STRESS
- STORAGE / GROUNDWATER DECLINE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CROP ECOSYSTEM LIVELIHOOD AND HEALTH EFFECTS; LAGS DIFFER → MONITOR EACH STAGE.

03

STAGE 03

DECLARATION DASHBOARD

DECLARATION DASHBOARD · RAINFALL + VEGETATIVE INDICES · CROP-SOWING PROGRESSION + SOIL MOISTURE · HYDROLOGICAL INDICES · GRADE STATUS · DO NOT INVENT ONE CUT-OFF

RAINFALL + VEGETATIVE INDICES

CROP-SOWING PROGRESSION + SOIL MOISTURE → HYDROLOGICAL INDICES → GRADE STATUS → DO NOT INVENT ONE CUT-OFF

HAZARD / RISK / RESILIENCE CONCEPT

- RAINFALL + VEGETATIVE INDICES
- CROP-SOWING PROGRESSION + SOIL MOISTURE

INSTITUTIONAL OR COMMUNITY RESPONSE

- HYDROLOGICAL INDICES
- GRADE STATUS → DO NOT INVENT ONE CUT-OFF

CAPACITY / STATUS / EVIDENCE LIMIT

- RAINFALL + VEGETATIVE INDICES
- CROP-SOWING PROGRESSION + SOIL MOISTURE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GRADE STATUS → DO NOT INVENT ONE CUT-OFF.

04

STAGE 04

HEAT HAZARD-TO-HARM CHAIN

HEAT HAZARD-TO-HARM CHAIN · IMD HAZARD CLASSIFICATION · HOT NIGHT · = HEAT STRESS AND UNEQUAL HARM

IMD HAZARD CLASSIFICATION

HUMIDITY / RADIATION / WIND / HOT NIGHT → WORK / HEALTH / HOUSING / WATER / COOLING → = HEAT STRESS AND UNEQUAL HARM

RISK CONDITION / INPUT

- IMD HAZARD CLASSIFICATION
- HUMIDITY / RADIATION / WIND / HOT NIGHT

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- WORK / HEALTH / HOUSING / WATER / COOLING
- = HEAT STRESS AND UNEQUAL HARM

LAST-MILE / INCLUSION / OUTCOME LIMIT

- IMD HAZARD CLASSIFICATION
- HUMIDITY / RADIATION / WIND / HOT NIGHT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WORK / HEALTH / HOUSING / WATER / COOLING.

05

STAGE 05

WET-BULB FIREWALL

WET-BULB FIREWALL · AMBIENT AIR TEMPERATURE · EVAPORATIVE-COOLING POTENTIAL · IMD HEAT WAVE · OPERATIONAL METEOROLOGICAL CRITERIA · NONE IS A UNIVERSAL PERSON-SAFE THRESHOLD

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
DRY-BULB → ambient air temperature	IMD HEAT WAVE → operational meteorological criteria	DRY-BULB → ambient air temperature
WET-BULB → evaporative-cooling potential	TRAP → NONE IS A UNIVERSAL PERSON-SAFE THRESHOLD	WET-BULB → evaporative-cooling potential

HAZARD / ACTOR / INSTRUMENT

- DRY-BULB → ambient air temperature
- WET-BULB → evaporative-cooling potential

DECISIVE RISK OR MANDATE DISTINCTION

- IMD HEAT WAVE → operational meteorological criteria
- TRAP → NONE IS A UNIVERSAL PERSON-SAFE THRESHOLD

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- DRY-BULB → ambient air temperature
- WET-BULB → evaporative-cooling potential

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → NONE IS A UNIVERSAL PERSON-SAFE THRESHOLD.

06

STAGE 06

HEAT ACTION PLAN RAIL

HEAT ACTION PLAN RAIL · 1 SEASONAL RISK AND RESPONSIBILITY MAP · 2 FORECAST · WARNING AND PUBLIC MESSAGE · 3 WATER COOLING WORK SCHOOL AND HEALTH ACTION · 4 REVIEW REACH EQUITY AND OUTCOMES

1 SEASONAL RISK AND RESPONSIBILITY MAP

2 FORECAST / WARNING AND PUBLIC MESSAGE → 3 WATER COOLING WORK SCHOOL AND HEALTH ACTION → 4 REVIEW REACH EQUITY AND OUTCOMES

HAZARD / RISK / RESILIENCE CONCEPT

- 1 SEASONAL RISK AND RESPONSIBILITY MAP
- 2 FORECAST / WARNING AND PUBLIC MESSAGE

INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 WATER COOLING WORK SCHOOL AND HEALTH ACTION
- 4 REVIEW REACH EQUITY AND OUTCOMES

CAPACITY / STATUS / EVIDENCE LIMIT

- 1 SEASONAL RISK AND RESPONSIBILITY MAP
- 2 FORECAST / WARNING AND PUBLIC MESSAGE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 WATER COOLING WORK SCHOOL AND HEALTH ACTION.

07

STAGE 07

WATER-AGRICULTURE PORTFOLIO

WATER-AGRICULTURE PORTFOLIO · DRINKING WATER + DEMAND MANAGEMENT · SURFACE WATER MONITORING · WATERSHED RECHARGE STORAGE GOVERNANCE · CROP LIVESTOCK LIVELIHOOD CONTINGENCY

DRINKING WATER + DEMAND MANAGEMENT

GROUND / SURFACE WATER MONITORING → WATERSHED RECHARGE STORAGE GOVERNANCE → CROP LIVESTOCK LIVELIHOOD CONTINGENCY

HAZARD / RISK / RESILIENCE CONCEPT

- DRINKING WATER + DEMAND MANAGEMENT
- GROUND / SURFACE WATER MONITORING

INSTITUTIONAL OR COMMUNITY RESPONSE

- WATERSHED RECHARGE STORAGE GOVERNANCE
- CROP LIVESTOCK LIVELIHOOD CONTINGENCY

CAPACITY / STATUS / EVIDENCE LIMIT

- DRINKING WATER + DEMAND MANAGEMENT
- GROUND / SURFACE WATER MONITORING

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATERSHED RECHARGE STORAGE GOVERNANCE; CROP LIVESTOCK LIVELIHOOD CONTINGENCY.

08

STAGE 08

URBAN-LABOUR-HEALTH MATRIX

URBAN-LABOUR-HEALTH MATRIX · BUILT FORM · COOL ROOF SHADE VENTILATION TREES · TIMING REST WATER COMMUNICATION · SURVEILLANCE TRIAGE REFERRAL CONTINUITY · POOR HOUSING INFORMAL WORK VULNERABLE GROUPS

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
BUILT FORM → cool roof shade ventilation trees	HEALTH → surveillance triage referral continuity	BUILT FORM → cool roof shade ventilation trees
LABOUR → timing rest water communication	ACCESS → poor housing informal work vulnerable groups	LABOUR → timing rest water communication

HAZARD / ACTOR / INSTRUMENT

- BUILT FORM → cool roof shade ventilation trees
- LABOUR → timing rest water communication

DECISIVE RISK OR MANDATE DISTINCTION

- HEALTH → surveillance triage referral continuity
- ACCESS → poor housing informal work vulnerable groups

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- BUILT FORM → cool roof shade ventilation trees
- LABOUR → timing rest water communication

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACCESS → poor housing informal work vulnerable groups.

09

STAGE 09

COMPOUND-RISK WEB

COMPOUND-RISK WEB · POWER PRESSURE · COOLING DEMAND · DRY FUEL · FIRE RISK · DEEPER LIVELIHOOD AND SERVICE STRESS · MUST REMEMBER · DROUGHT AND HEAT ARE SLOW-ONSET OR COMPOUNDING RISKS SHAPED

DROUGHT → WATER / CROP / POWER PRESSURE → HEAT → HEALTH / LABOUR / COOLING DEMAND → DRY FUEL → FIRE RISK → FEEDBACK → DEEPER LIVELIHOOD AND SERVICE STRESS → MUST REMEMBER: Drought and heat are slow-onset or compounding risks shaped by

HAZARD / RISK / RESILIENCE CONCEPT

- DROUGHT → WATER / CROP / POWER PRESSURE
- HEAT → HEALTH / LABOUR / COOLING DEMAND

INSTITUTIONAL OR COMMUNITY RESPONSE

- DRY FUEL → FIRE RISK
- FEEDBACK → DEEPER LIVELIHOOD AND SERVICE STRESS

CAPACITY / STATUS / EVIDENCE LIMIT

- MUST REMEMBER: Drought and heat are slow-onset or compounding risks shaped by...
- DROUGHT → WATER / CROP / POWER PRESSURE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Drought and heat are slow-onset or compounding risks shaped by.

10

STAGE 10

EARLY ACTION OR MALADAPTATION

EARLY ACTION OR MALADAPTATION · FORECAST + VULNERABILITY SIGNAL? · EARLY WATER HEALTH LABOUR LIVELIHOOD ACTION · ENERGY WATER LAND ACCESS MAINTENANCE · FAIL TEST · TRANSFERRED RISK · CLOSE DISTINCTION · METEOROLOGICAL DROUGHT IS NOT AGRICULTURAL OR HYDROLOGICAL DROUGHT

FORECAST + VULNERABILITY SIGNAL?

YES → EARLY WATER HEALTH LABOUR LIVELIHOOD ACTION → TEST → ENERGY WATER LAND ACCESS MAINTENANCE → FAIL TEST → MALADAPTATION / TRANSFERRED RISK

HAZARD / RISK / RESILIENCE CONCEPT

- FORECAST + VULNERABILITY SIGNAL?
- YES → EARLY WATER HEALTH LABOUR LIVELIHOOD ACTION

INSTITUTIONAL OR COMMUNITY RESPONSE

- TEST → ENERGY WATER LAND ACCESS MAINTENANCE
- FAIL TEST → MALADAPTATION / TRANSFERRED RISK

CAPACITY / STATUS / EVIDENCE LIMIT

- CLOSE DISTINCTION: Meteorological drought is not agricultural or hydrological drought;...
- FORECAST + VULNERABILITY SIGNAL?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Meteorological drought is not agricultural or hydrological drought;.

11

STAGE 11

SLOW-ONSET ANSWER SPINE

SLOW-ONSET ANSWER SPINE · DISTINGUISH ARIDITY SCARCITY AND DROUGHT TYPES · TRACE LAGS + MONITOR MULTIPLE INDICATORS · SEPARATE HEAT HAZARD FROM STRESS AND EXPOSURE · DESIGN EARLY ACTION + TEST EQUITY MALADAPTATION OUTCOME · EVIDENCE LIMIT · PRESERVE IMD THRESHOLD, STATION/REGION

DISTINGUISH ARIDITY SCARCITY AND DROUGHT TYPES → TRACE LAGS + MONITOR MULTIPLE INDICATORS → SEPARATE HEAT HAZARD FROM STRESS AND EXPOSURE → DESIGN EARLY ACTION + TEST EQUITY MALADAPTATION OUTCOME → EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Preserve IMD threshold, station/region

HAZARD / RISK / RESILIENCE CONCEPT

- DISTINGUISH ARIDITY SCARCITY AND DROUGHT TYPES
- TRACE LAGS + MONITOR MULTIPLE INDICATORS

INSTITUTIONAL OR COMMUNITY RESPONSE

- SEPARATE HEAT HAZARD FROM STRESS AND EXPOSURE
- DESIGN EARLY ACTION + TEST EQUITY MALADAPTATION OUTCOME

CAPACITY / STATUS / EVIDENCE LIMIT

- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Preserve IMD threshold, station/region;...
- DISTINGUISH ARIDITY SCARCITY AND DROUGHT TYPES

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Preserve IMD threshold, station/region;...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Preserve IMD threshold, station/region;.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · HEAT-WAVE LOCAL-STATUS STATEMENT AS DOCUMENT-PERIOD EVIDENCE · COLD WAVE'S EXPLICIT NON-NATIONAL-PLAN DESIGN · VISIONIAS STATES · DROUGHT'S MULTI-INDICATOR DECLARATION PROCESS · BUNDELKHAND MENTHA CULTIVATION AS A CONCRETE POLITICAL-ECONOMY · CWMI AND THE FALKENMARK INDEX AS TWO DISTINCT MEASUREMENT · VISIONIAS'S OWN CRITIQUE OF HEAT-WAVE MANAGEMENT NAMES "LACK OF

OPTIONAL RISK / RESILIENCE DEPTH

- Heat-wave local-status statement as document-period evidence :
- Cold wave's explicit non-national-plan design : VisionIAS states
- CAUTION: Drought's multi-indicator declaration process : drought-affected
- Bundelkhand Mentha cultivation as a concrete political-economy

MANDATE, WARNING OR OUTCOME LIMIT

- CWMI and the Falkenmark Index as two distinct measurement
- CAUTION: VisionIAS's own critique of heat-wave management names "lack of
- CAUTION: Anticipatory action potential : because drought is explicitly a
- VisionIAS's HAP community-outreach material names specific

COMPARATIVE HAZARD / GOVERNANCE NUANCE

- CAUTION: Drought-linked distress migration and heat-linked occupational
- CAUTION: Short-term relief versus long-term structural adaptation is the
- CAUTION: Water-intensive cropping versus farmer income : discouraging
- NOT: The "a Heat Action Plan exists, therefore heat risk is managed"

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels